

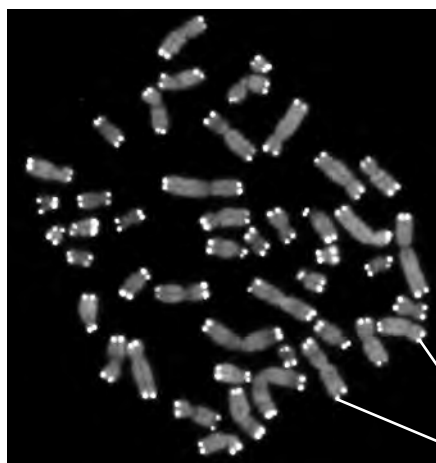
6 A telomere is a repetitive nucleotide sequence at the end of a chromosome.

The telomere protects the end of each chromosome from damage.

Each time DNA is synthesised the telomeres become shorter.

When the telomeres become too short the cell dies.

The photograph shows telomeres on the ends of chromosomes.



telomeres

(Source: © Hesus Padilla-Nash and Thomas Ried, National Cancer Institute, National Institutes Of Health / Science Photo Library)

(a) (i) Which diagram shows a mononucleotide of DNA?

(1)

Key



Sugar



Phosphate

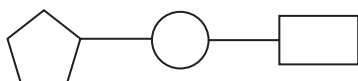


Base

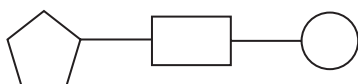
— Bond



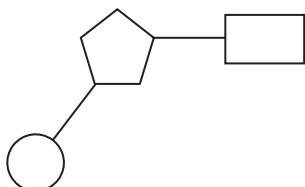
A



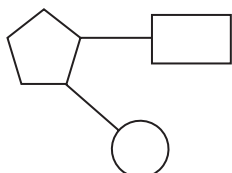
B



C

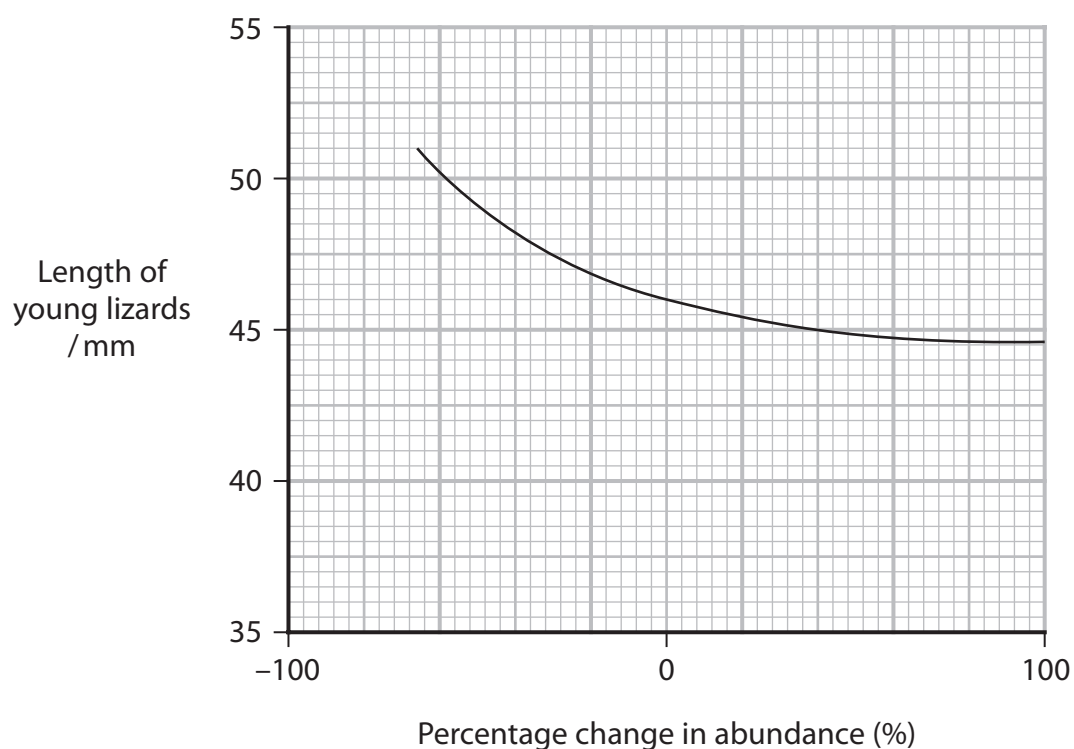


D



- (ii) The scientists used the data to predict a relationship between the length of the young lizards and the percentage change in abundance of lizards over the 12 year period.

The graph shows this relationship.



Describe the relationship between percentage change in abundance of lizards and length of young lizards.

(2)

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- (iii) These scientists also looked at the length of the telomeres in the cells of these young lizards.

Explain why the abundance of the longer lizards is affected by the number of cell divisions taking place as they grow.

Use all the information in the question to support your answer.

(3)

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(Total for Question 6 = 10 marks)